

SIMATS ENGINEERING

ASSESSMENT TOOL III – VISUALIZATION CRITIQUE & REDESIGN

Course: Data Handling and Visualization	Code: DSA0601	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No : Name:

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Visualization Critique & Redesign

Weightage: 20%

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

Criteria	Evaluation of Existing Visualization 25%	Redesign Strategy 25%	Application of Visualization Principles 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							

Q4							
Q5							

AT3 – Visualization Critique & Redesign

R Programming Assessment – Distribution and Proportion Visualization

Instructions: For each question, use R/RStudio to reproduce the given visualization, critique its effectiveness, identify misleading or inappropriate visual encodings, and redesign it using a more suitable visualization. Include the R code, redesigned visualization, and a concise justification of your design decisions.

Question 1 – Critique and Redesign an ECDF and Q-Q Plot

Dataset: Salary_Distribution.csv

Dataset:

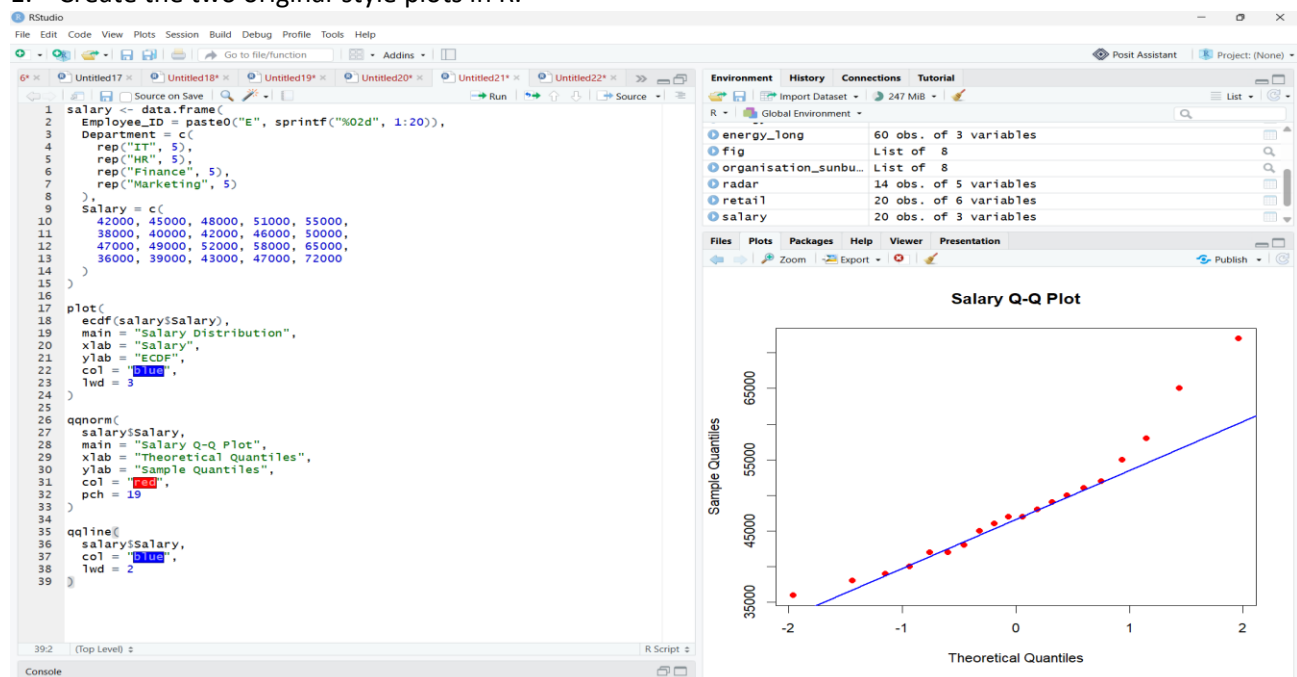
Employee_ID	Department	Salary
E01	IT	42000
E02	IT	45000
E03	IT	48000
E04	IT	51000
E05	IT	55000
E06	HR	38000
E07	HR	40000
E08	HR	42000
E09	HR	46000
E10	HR	50000
E11	Finance	47000
E12	Finance	49000
E13	Finance	52000

E14	Finance	58000
E15	Finance	65000
E16	Marketing	36000
E17	Marketing	39000
E18	Marketing	43000
E19	Marketing	47000
E20	Marketing	72000

Assessment Task

A salary analyst has created an ECDF and a Q-Q plot to study employee salary distribution, but the charts use unclear labels, inappropriate scaling, and excessive decoration.

1. Create the two original-style plots in R.



2. Identify at least three visualization problems.

Answer:

Problem 1 – Unclear labels

The original visualization uses unclear labels for the salary values and cumulative distribution, making interpretation difficult.

Problem 2 – Inappropriate scaling

The salary axis does not provide an easily interpretable scale for comparing the minimum, middle, and maximum salaries.

Problem 3 – Excessive decoration

Unnecessary colors and visual decoration can distract the viewer from the actual salary distribution.

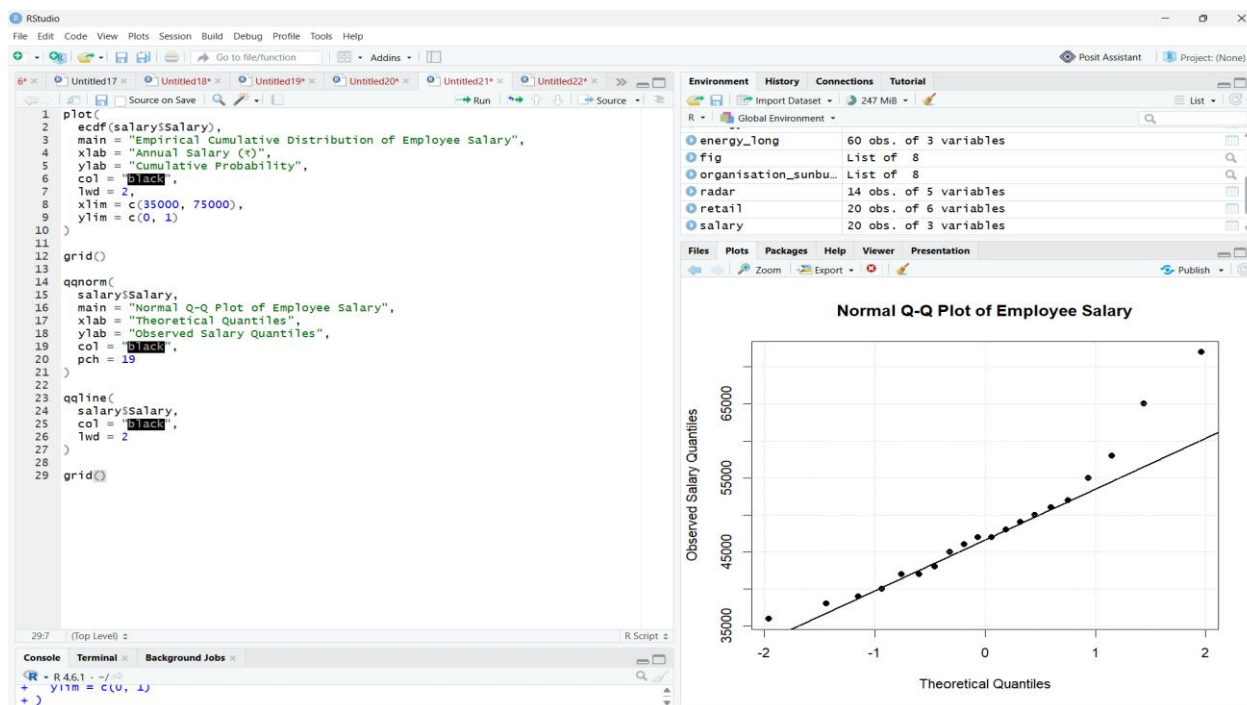
Problem 4 – Poor reference information

The Q-Q plot needs a clear reference line so that deviations from normality can be identified easily.

Problem 5 – Difficult outlier identification

The high salaries, especially ₹65,000 and ₹72,000, may not be immediately noticeable without a suitable scale and clean presentation.

3. Redesign the plots using appropriate scales, labels, titles, reference lines, and minimal visual clutter.



4. Explain how the redesigned plots improve interpretation of skewness, outliers, and normality.

Answer:

Skewness

The redesigned ECDF provides a clearer view of how salaries accumulate across the salary range. The relatively high values at ₹65,000 and ₹72,000 indicate a longer upper tail, suggesting some positive/right skewness.

Outliers

The values **₹65,000** and **₹72,000** are considerably higher than many of the other salaries. The improved salary scale makes these extreme observations easier to identify.

Normality

The redesigned Q-Q plot contains a clear reference line. If the observed points lie close to this

line, the distribution is approximately normal. Systematic deviations from the line indicate departures from normality, particularly in the tails.

Overall improvement

The redesign improves interpretation by providing:

- Clear axis labels.
- Appropriate salary scale.
- Meaningful cumulative probability scale.
- Clear Q-Q reference line.
- Reduced visual clutter.
- Better identification of extreme values.

5. State whether a Q-Q plot alone is sufficient to conclude that the data are normally distributed.

Answer:

No. A Q-Q plot alone is not sufficient.

A Q-Q plot is a graphical method for assessing whether the data approximately follow a normal distribution. However, it should be supported by other methods such as:

- Histogram.
- Density plot.
- Skewness and kurtosis.
- Shapiro-Wilk normality test.
- Descriptive statistical analysis.

Therefore, a Q-Q plot should be combined with statistical tests and other visualizations before making a final conclusion about normality.

Question 2 – Critique and Redesign Many Distributions Using Bean

Plots Dataset: Student_Scores.csv

Dataset:

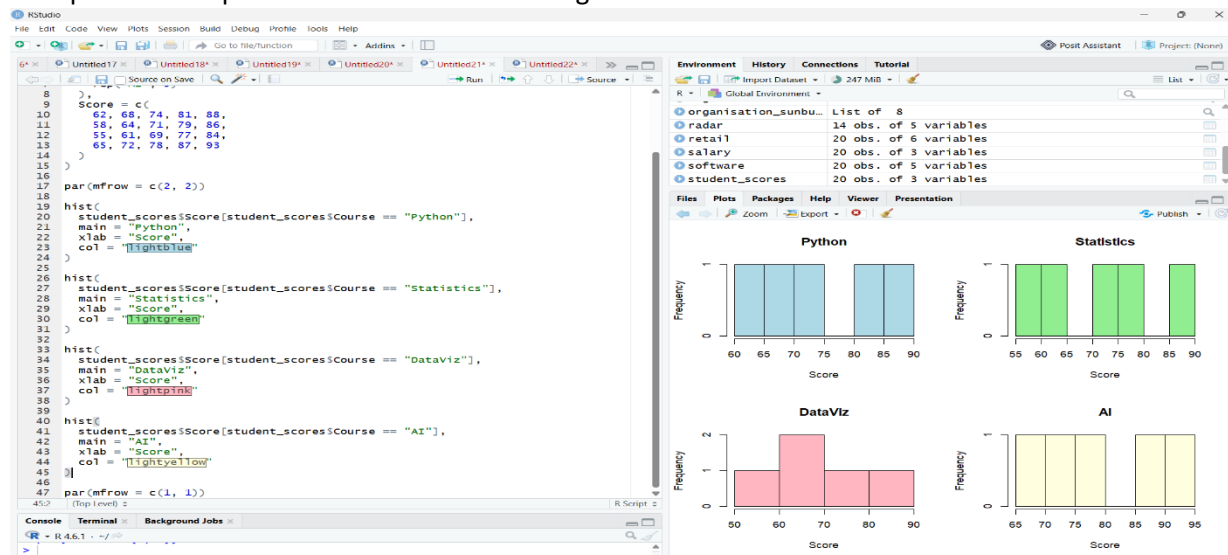
Student_ID	Course	Score
S01	Python	62
S02	Python	68
S03	Python	74
S04	Python	81
S05	Python	88

S06	Statistics	58
S07	Statistics	64
S08	Statistics	71
S09	Statistics	79
S10	Statistics	86
S11	DataViz	55
S12	DataViz	61
S13	DataViz	69
S14	DataViz	77
S15	DataViz	84
S16	AI	65
S17	AI	72
S18	AI	78
S19	AI	87
S20	AI	93

Assessment Task

A college report uses four separate histograms with different bin widths to compare course scores. The comparison is difficult because the scales and layouts are inconsistent.

1. Reproduce the problematic visualization using R.



2. Critique the use of separate histograms, inconsistent scales, and bin widths.

Answer:

Problem 1 – Separate histograms

Using four separate histograms makes direct comparison between courses difficult.

Problem 2 – Inconsistent scales

If the x-axis or y-axis scales differ between histograms, the distributions can appear more different than they actually are.

Problem 3 – Different bin widths

Different bin widths can change the visual appearance of a distribution. A course may appear more concentrated or more spread out simply because of the chosen bin width.

Problem 4 – Difficult comparison

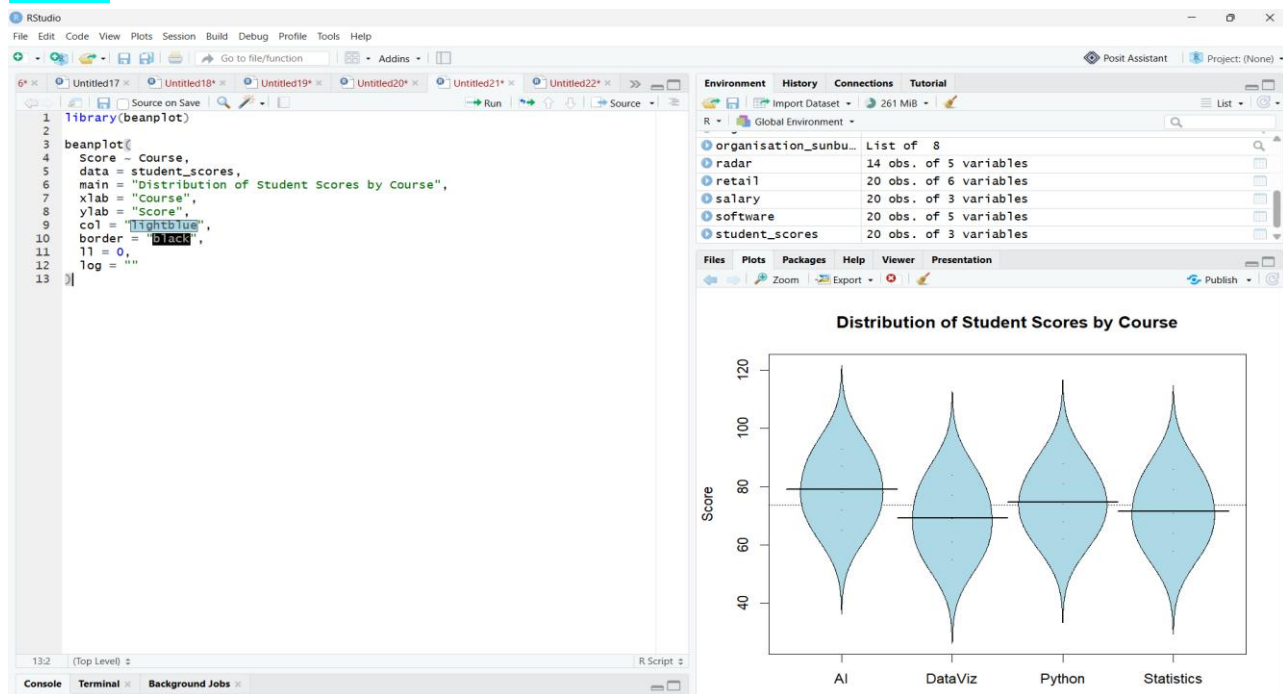
The viewer has to move between four different plots to compare the courses.

Problem 5 – Poor representation of small datasets

Each course contains only five scores. Histograms are not particularly effective for showing the detailed distribution of such small groups.

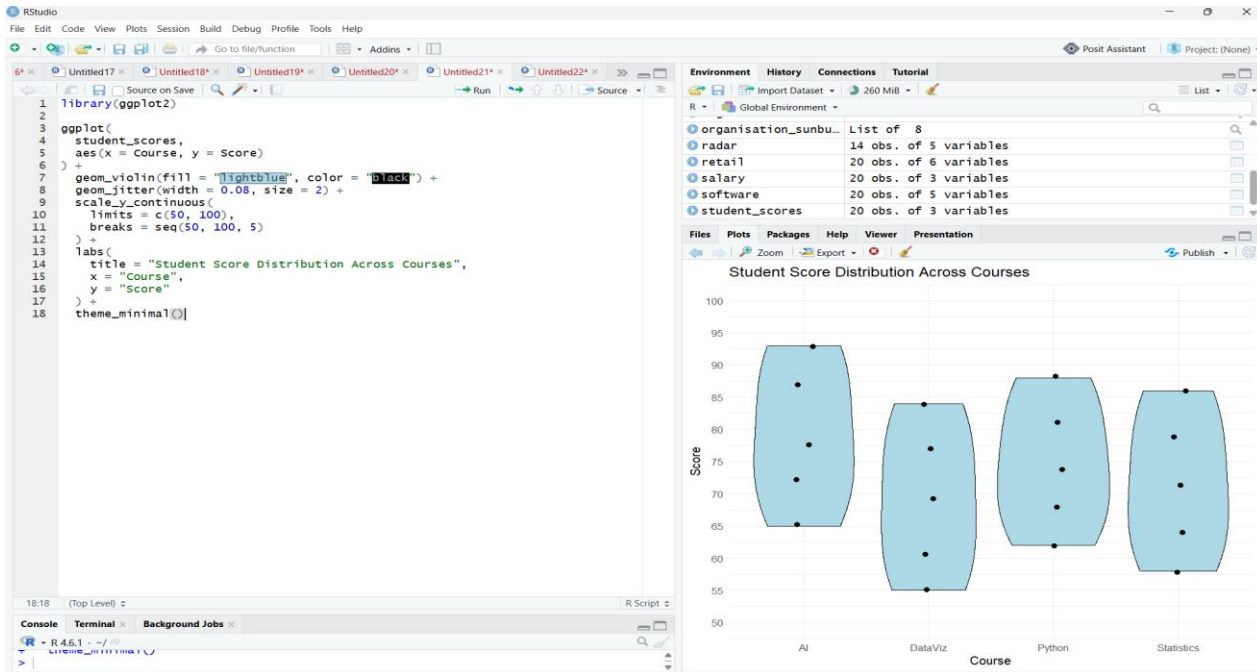
3. Redesign the visualization using bean plots or a suitable violin/density-based alternative.

Answer:



4. Add an appropriate common scale and clear course labels.

Answer:



5. Explain which course has the highest concentration of scores and which course shows the greatest spread.

Answer:

Highest concentration

Python has scores:

62, 68, 74, 81, 88

Range : 88-62=26

Statistics: 86-58=28

DataViz: 84-55=29

AI: 93-65=28

Therefore, Python has the highest concentration of scores, because it has the smallest range of 26 points.

Greatest spread

DataViz has the greatest spread, because its score range is: 84-55=29

Therefore:

- Highest concentration: Python
- Greatest spread: DataViz

Question 3 – Critique and Redesign Proportion Visualizations

Dataset: Regional_Purchases.csv

Dataset:

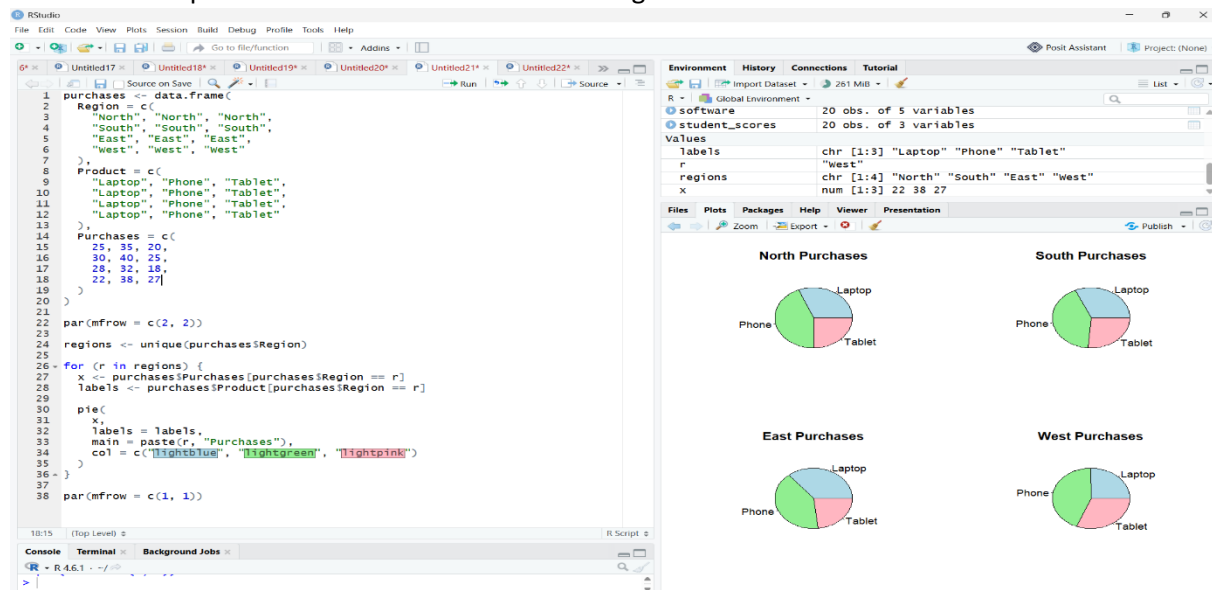
Region	Product	Purchases
North	Laptop	25
North	Phone	35

North	Tablet	20
South	Laptop	30
South	Phone	40
South	Tablet	25
East	Laptop	28
East	Phone	32
East	Tablet	18
West	Laptop	22
West	Phone	38
West	Tablet	27

Assessment Task

A manager presents product proportions using four pie charts, one for each region. The charts use different starting angles, 3-D effects, and inconsistent labeling.

1. Create a representative version of the misleading visualization.



2. Identify at least four design problems.

Answer:

Problem 1 – Use of pie charts

Four separate pie charts make it difficult to compare the proportions of products across regions.

Problem 2 – Different starting angles

Different starting angles make similar-sized sectors appear different and make comparisons more difficult.

Problem 3 – 3-D effects

Three-dimensional effects can distort the apparent size of sectors. A sector closer to the viewer may look larger than it actually is.

Problem 4 – Inconsistent labeling

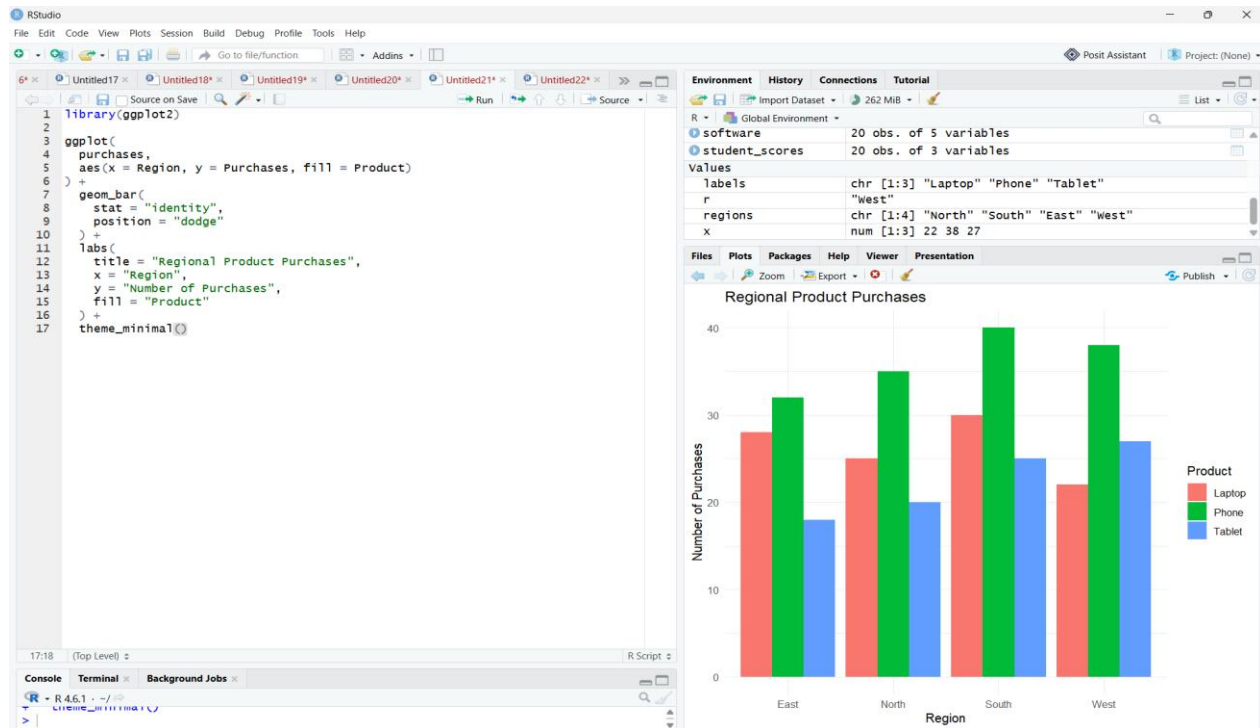
Inconsistent labels make it difficult to identify and compare the products between regions.

Problem 5 – Difficult comparison

Comparing angles and areas of pie slices across four separate charts is harder than comparing aligned bars.

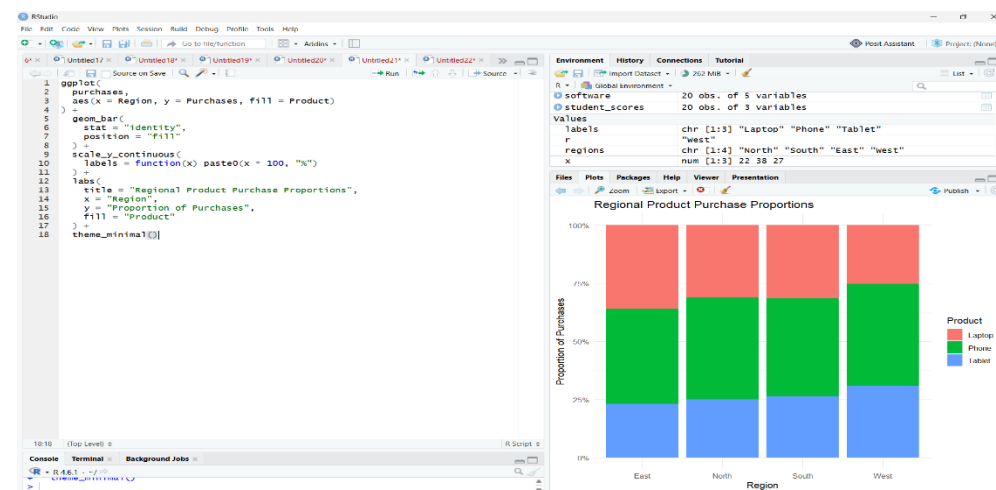
3. Redesign the analysis using side-by-side bars and a 100% stacked bar chart.

Answer: Side-by-side bar chart



The side-by-side bar chart allows the purchase values for Laptop, Phone, and Tablet to be compared directly within and across regions.

B. 100% stacked bar chart



4. Compare the two redesigns and determine which is better for comparing regional proportions.

Answer:

Side-by-side bar chart

Advantages:

- Easy to compare actual purchase quantities.
- Product values can be compared directly.
- Common baseline improves accuracy.
- Differences between products are easy to see.

100% stacked bar chart

Advantages:

- Shows the proportion of each product within a region.
- Every region has the same 100% total.
- Useful for comparing regional composition.
- Clearly communicates percentage differences.

5. Explain why 3-D effects and inconsistent scales can lead to misleading conclusions.

Answer:

3-D effects

3-D pie charts introduce perspective distortion. Sectors located in the foreground can appear larger than sectors of the same actual size in the background.

Inconsistent scales

When different charts use inconsistent scales, the same numerical difference may appear visually larger or smaller. This can cause the viewer to draw incorrect conclusions.

Question 4 – Critique and Redesign Nested Proportions

Dataset: Sales_Hierarchy.csv

Dataset:

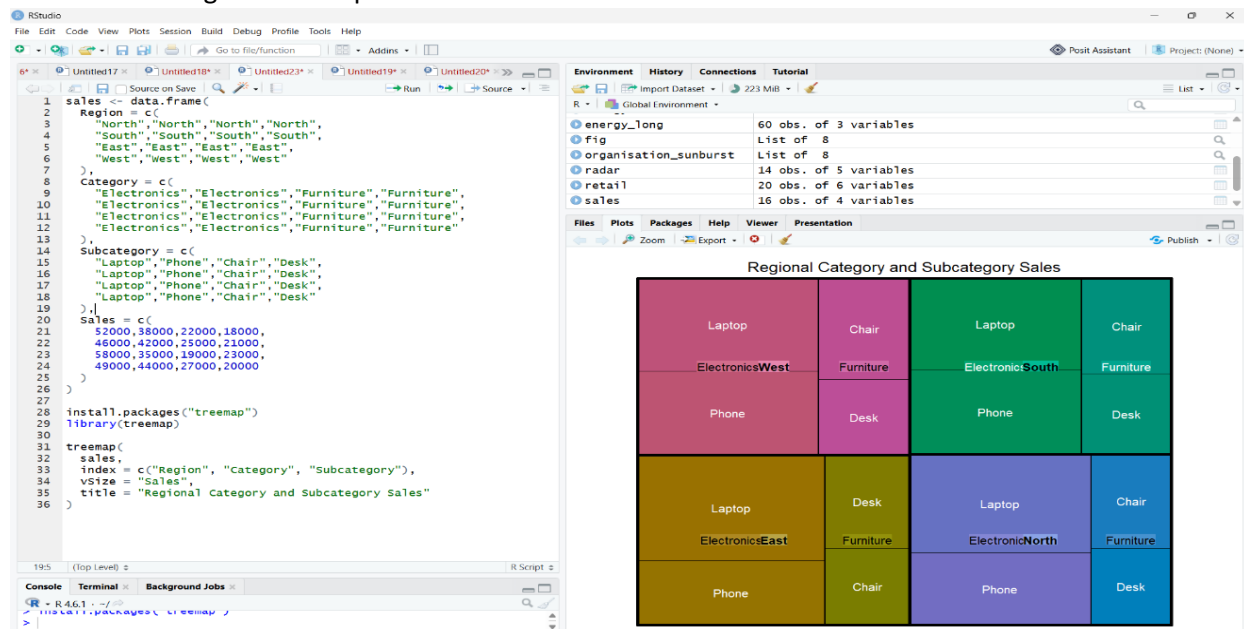
Region	Category	Subcategory	Sales
North	Electronics	Laptop	52000
North	Electronics	Phone	38000
North	Furniture	Chair	22000
North	Furniture	Desk	18000
South	Electronics	Laptop	46000
South	Electronics	Phone	42000
South	Furniture	Chair	25000

South	Furniture	Desk	21000
East	Electronics	Laptop	58000
East	Electronics	Phone	35000
East	Furniture	Chair	19000
East	Furniture	Desk	23000
West	Electronics	Laptop	49000
West	Electronics	Phone	44000
West	Furniture	Chair	27000
West	Furniture	Desk	20000

Assessment Task

An organization uses a treemap to communicate Region → Category → Subcategory sales. Small areas have unreadable labels and the color scale does not clearly communicate the hierarchy.

1. Create the original treemap.



2. Critique the size, labeling, color, and hierarchy encodings.

Answer:

Size

The treemap uses area to represent sales. Large areas represent high sales, but comparing areas precisely is difficult.

Labeling

Small rectangles have limited space, so their labels may become difficult to read or overlap.

Color

If similar colors are used for different categories, it becomes difficult to distinguish Electronics from Furniture.

Hierarchy

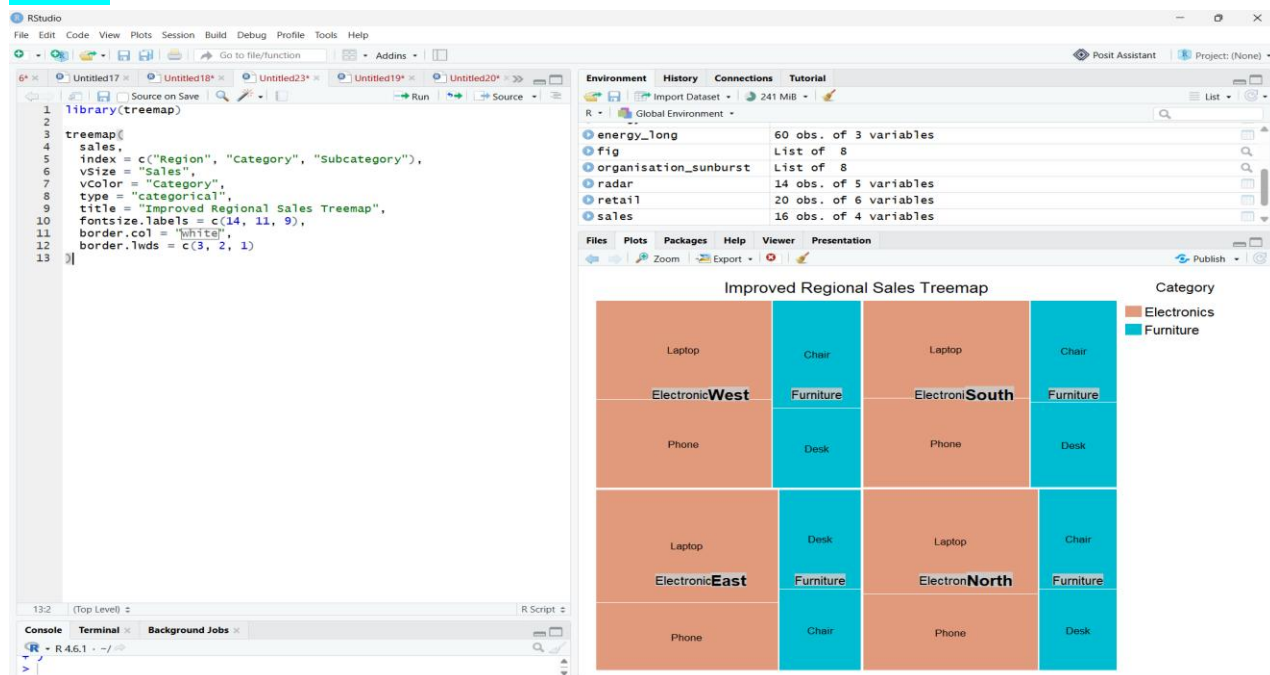
The Region → Category → Subcategory hierarchy is represented through nested rectangles, but the hierarchy may not be immediately clear when many small areas are present.

Overall problems

1. Small areas have unreadable labels.
2. Similar colors make categories difficult to distinguish.
3. Area comparison is less precise than bar-length comparison.
4. The hierarchy is not immediately obvious.
5. Too much information can make the visualization visually crowded.

3. Redesign the visualization as an improved treemap.

Answer:



4. Create a sunburst alternative and compare it with the redesigned treemap.

Answer:

Treemap

Advantages:

- Efficient use of rectangular space.
- Easy to see the largest sales groups.
- Suitable for showing hierarchical proportions.
- Works well when many categories are present.

Disadvantages:

- Small rectangles can have unreadable labels.

- Comparing areas is difficult.
- Deep hierarchies can become crowded.

Sunburst

Advantages:

- Clearly shows multiple hierarchical levels.
- Circular layers visually represent Region → Category → Subcategory.
- Useful for understanding hierarchical structure.

Disadvantages:

- Comparing segment sizes is more difficult.
- Small segments can be difficult to read.
- Less suitable for precise sales comparisons.

5. Recommend one visualization for a management presentation and justify your choice.

Answer:

Recommended visualization: Improved Treemap

The **improved treemap** is recommended for a management presentation because it provides a compact overview of the entire **Region → Category → Subcategory → Sales** hierarchy.

It allows managers to quickly identify:

- Regions with high sales.
- Categories with high sales.
- High-performing subcategories.
- Relative sales contribution.
- Differences between large and small sales groups.

The improved treemap is therefore more suitable for a **management-level summary**, while the sunburst can be used as an alternative when the main objective is to emphasize the hierarchical structure

Question 5 – Critique and Redesign Flow-Based Proportions

Dataset: Career_Flow.csv

Dataset:

Specialization	Internship	Placement_Sector	Students
Data Analytics	Yes	IT	18
Data Analytics	Yes	Finance	12
Data Analytics	No	IT	8
AI	Yes	IT	20
AI	Yes	Healthcare	10
AI	No	IT	6

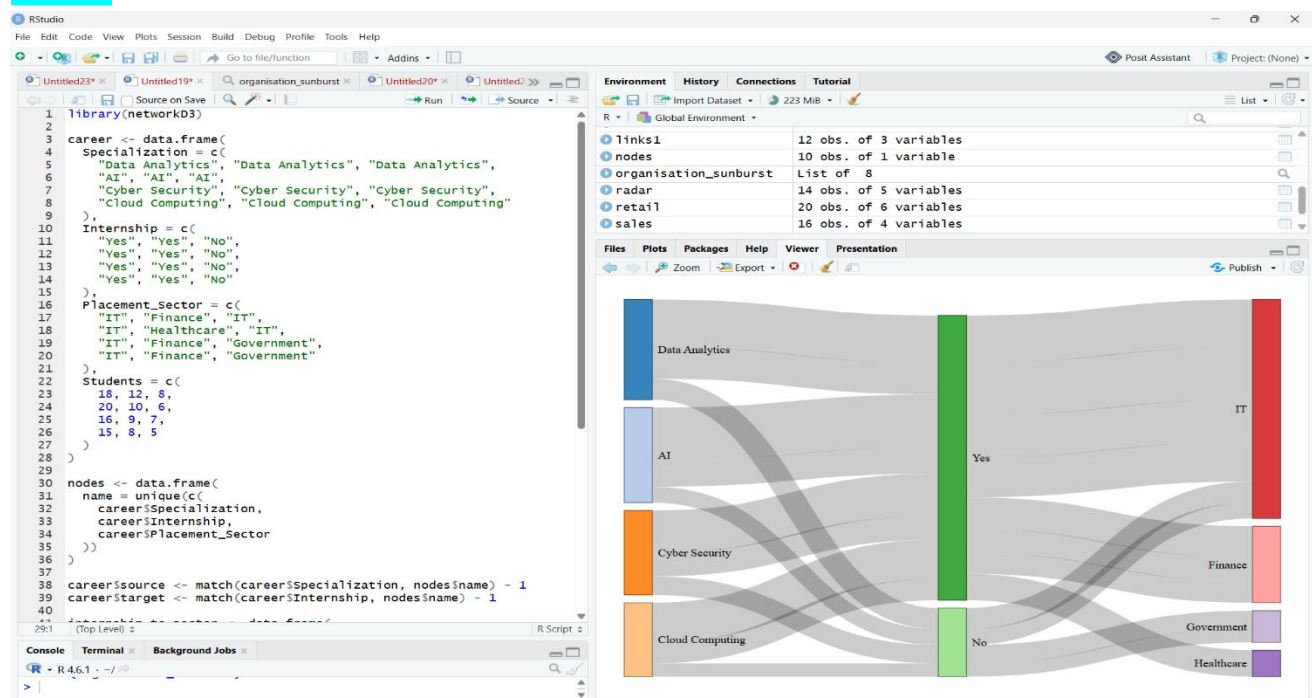
Cyber Security	Yes	IT	16
Cyber Security	Yes	Finance	9
Cyber Security	No	Government	7
Cloud Computing	Yes	IT	15
Cloud Computing	Yes	Finance	8
Cloud Computing	No	Government	5

Assessment Task

A placement office uses a Sankey diagram to show Specialization → Internship → Placement Sector. The original chart uses excessively similar colors, unlabelled flows, and proportions that are difficult to compare.

1. Create a Sankey diagram representing the data.

Answer:



2. Identify at least four potential sources of confusion or misleading interpretation.

Answer:

Problem 1 – Similar colors

If different flows use very similar colors, it becomes difficult to distinguish between specialization, internship status, and placement sectors.

Problem 2 – Unlabelled flows

Without clear labels, users may not know the number of students represented by each flow.

Problem 3 – Difficult proportion comparison

The widths of multiple flows can be difficult to compare accurately, particularly when several flows have similar sizes.

Problem 4 – Excessive visual complexity

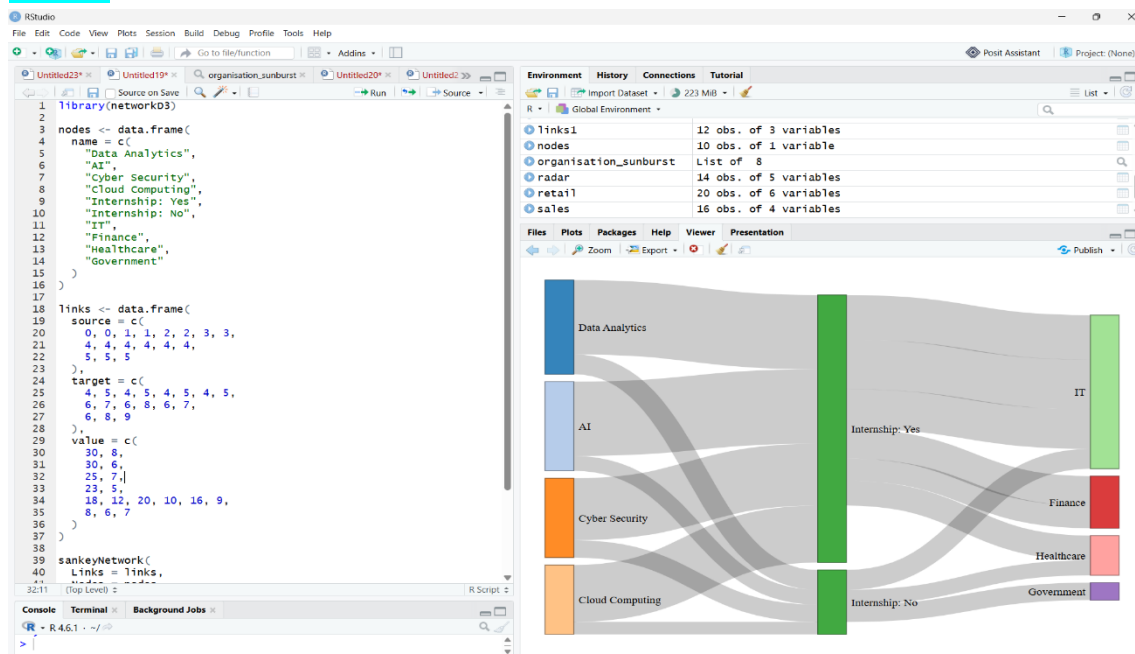
Multiple paths crossing or connecting different nodes can make the diagram difficult to follow.

Problem 5 – Weak visual hierarchy

If all nodes and flows have similar visual importance, the major student pathways do not stand out clearly.

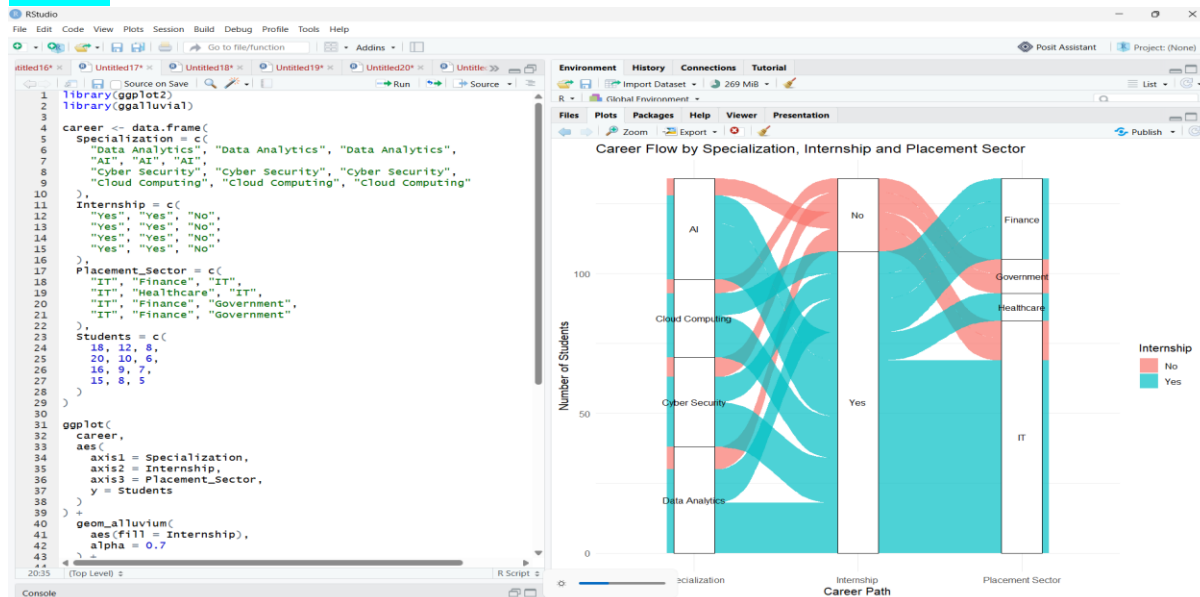
3. Redesign the Sankey diagram with meaningful labels, consistent flow representation, and an appropriate visual hierarchy.

Answer:



4. Create a parallel sets alternative and compare the two designs.

Answer:



5. Explain which visualization better communicates the major student flows and why.

Answer:

Recommended visualization: Sankey Diagram

The redesigned Sankey diagram better communicates the major student flows because the width of each flow directly represents the number of students.

The largest flow is:

Data Analytics → Internship Yes=30

These major flows can be identified quickly from the Sankey diagram.

The Sankey design also clearly presents the progression:

Therefore, Sankey is the better visualization for communicating the major student flows, while Parallel Sets is a useful alternative for examining relationships among several categorical variables.

AT3 Expected Student Output

- R program/script used to create and redesign the visualization.
- Original/problematic visualization reproduced or clearly simulated.
- Written critique identifying visualization and design problems.
- Redesigned visualization with improved visual encodings.
- Brief justification explaining why the redesign is more accurate, readable, and less misleading. •

Final recommendation where comparison of alternative visualizations is requested.

Suggested AT3 Assessment Rubric

Criterion	Weight
Identification of visualization problems	20%
Correct R programming and data preparation	20%
Quality and appropriateness of redesign	30%
Justification and visualization critique	20%
Clarity, labeling, and presentation	10%